

Comprehensive literature review database for ML-based stunting prediction using 2024 MDHS

This structured literature review covers 100+ papers across seven domains, providing the citation evidence base for a journal paper titled "Survey-weighted machine learning for nationally representative prediction of child stunting in Malawi." The review identifies a critical methodological gap: ~90% of ML studies using DHS data ignore survey sampling weights, inflating performance metrics and producing non-representative predictions. The proposed study addresses this gap using the 2024 Malawi DHS — the first application of survey-weighted, multi-algorithm ML to predict stunting with nationally representative data in Malawi.

1. Prevalence and burden of child stunting in Malawi and sub-Saharan Africa

1.1 Malawi stunting prevalence and trends (DHS series 1992-2024)

Malawi has achieved a **historic decline in child stunting from 55% in 1992 to approximately 30% in 2024**, representing the first time prevalence has dropped below the symbolic 30% threshold. This trajectory, documented across six successive DHS rounds, reflects decades of sustained policy investment yet leaves nearly one in three Malawian children chronically malnourished.

Key citations and data points:

- **National Statistical Office [Malawi] and ICF (2025).** *Malawi Demographic and Health Survey 2024: Final Report*. Zomba, Malawi, and Rockville, Maryland, USA: NSO, Utica International, and ICF. [Report FR398]
 - Sixth DHS round (preceded by 1992, 2000, 2004, 2010, 2015-16). (Nsomalawi) (Dcafs-tipdep-donors-mw) Fieldwork May 13-August 31, 2024. (Jointdatacenter) Sample: ~23,000 households across 28 districts and 4 cities (Nsomalawi) plus Dzaleka refugee camp. **Stunting prevalence: ~30%**; wasting ~2%. Under-five mortality dropped to **49 per 1,000 live births** (from 234 in 1992). Vaccination coverage (12-23 months): 84%. Skilled birth attendance: 92%. (Nyasa Times)
 - **Relevance:** Primary dataset for the ML prediction study. Establishes outcome variable distribution and sampling frame.
- **Doctor, H.V. & Nkhana-Salimu, S. (2017).** "Trends and Determinants of Child Growth Indicators in Malawi." *AIMS Public Health*, 4(6):590-614.
 - Analysis of 31,630 children across five DHS rounds (1992-2015/16). (PubMed) Stunting trajectory: **55.6% (1992) → 48% (2000) → 48% (2004) → 47% (2010) → 32.6% (2015/16)**. Males, children >6 months, rural residents, and central/southern region residents consistently higher risk. (ResearchGate)
 - **Relevance:** Establishes the long-term trend context. Demonstrates persistent geographic and demographic differentials ML models should capture.
- **Kumchulesi, G. (2020).** "Explaining the Decline in Child Stunting in Malawi between 2010 and 2015." *AERC Policy Brief No. 711*.
 - Stunting dropped from 47% (2010) to 37% (2015/16) — a ~10 percentage point decline. Blinder-Oaxaca decomposition explained only **5% of the decline**, attributable primarily to improved household wealth. 95% of the decline remained unexplained. (Aercafrica)
 - **Relevance:** The large unexplained component directly motivates using ML methods capable of detecting complex, non-linear interactions among predictors that traditional decomposition methods miss.
- **Chirwa, E.W. & Ngalawa, H. (2018).** "Trends and Determinants of Child Growth Indicators in Malawi and Implications for the Sustainable Development Goals." *Annals of Global*

- Comprehensive trend analysis showing average annual stunting reduction of **1.0 percentage points** over 1992–2015/16. (PubMed Central) Male children and central/southern regions consistently most affected.

1.2 Regional comparisons across sub-Saharan Africa

- **Pooled SSA prevalence (Bayesian multilevel analysis, 2022).** "Pooled prevalence of stunting and associated factors among children aged 6–59 months in Sub-Saharan Africa countries." *PLOS ONE*. DOI: 10.1371/journal.pone.0275889.
 - 173,483 weighted samples from 35 SSA countries (DHS 2008–2020). Pooled stunting: **35% (95% CI: 34.87–35.31)**. East Africa highest at **37%**. Key predictors: male sex (AOR=1.27), small birth size (AOR=1.29), no maternal education (AOR=3.07). (PLOS)
 - *Relevance:* Establishes SSA baseline for contextualizing Malawi's burden; identifies cross-country consistent predictors for ML feature selection.
- **Wand, H. et al. (2025).** "Population-Level Impact of Risk Factors on Stunting Among Children Under Five in Sub-Saharan Africa (2015–2022)." *Acta Paediatrica*. DOI: 10.1111/apa.70161.
 - 179,572 children across 24 SSA countries. Overall SSA decline from 41% (2016) to 20% (2023). **Malawi (35%), Madagascar (39%), and Burundi (55%)** among the highest. Lack of basic household amenities accounted for **24–57% of stunting cases**. (Wiley Online Library)
- **Quamme, S.H. & Iversen, P.O. (2022).** "Prevalence of child stunting in Sub-Saharan Africa and its risk factors." *Clinical Nutrition Open Science*, 42:49–61. DOI: 10.1016/j.nutos.2022.01.009.
 - Narrative review of 13 articles (2000–2020). Average SSA prevalence **41%**. Higher in children >1 year, boys, low birth weight, shorter interpregnancy intervals. (ScienceDirect)

1.3 Global burden and SDG progress

- **UNICEF, WHO, World Bank Group (2025).** *Joint Child Malnutrition Estimates: Key Findings of the 2025 Edition*.
 - **150.2 million** children under 5 stunted globally (**23.2% prevalence**). (UNSD) 43% in Africa. (WHO) Boys: 24.4% vs. Girls: 21.9%. Only **1 in 4 countries** on track to halve stunting by 2030. Target will be missed by **46.0 million children** at current rates. (WHO)
- **Victora, C.G. et al. (2021).** "Revisiting maternal and child undernutrition in low-income and middle-income countries." *The Lancet*, 397(10282):1388–1399.
 - Global childhood stunting fell from 32.5% (2000) to 21.9% (2017). (Global Nutrition Report) Progress more pronounced in middle-income than low-income countries. (ENN) 4.7% simultaneously stunted and wasted (4.8× mortality increase). (The Lancet)
- **Black, R.E. et al. (2008).** "Maternal and child undernutrition: global and regional exposures and health consequences." *The Lancet*, 371(9608):243–260.
 - Foundational paper. Stunting, severe wasting, and IUGR responsible for **2.2 million child deaths** and **21% of DALYs** for children under 5. (Cesni-biblioteca)
- **Galasso, E. & Wagstaff, A. (2019).** "The aggregate income losses from childhood stunting." World Bank Policy Research Working Paper 8536. *Economics & Human Biology*.
 - Average per capita income penalty from stunting: **~7% globally, ~9–10% of GDP** in Africa/South Asia. (World Bank) Benefit-cost ratio of nutrition interventions: **5–6:1**. (ScienceDirect)

1.4 National nutrition policy context

- **Government of Malawi (2025).** *National Multisector Nutrition Strategic Plan 2025–2030*. Ministry of Health.
 - Aligned with MW2063, SUN 3.0, and WHA targets. [Health](#) Malnutrition costs ~US\$600 million annually (10.3% of GDP). [World Vision](#) Creates explicit demand for data-driven targeting tools.
- **Scaling Up Nutrition (2021).** SUN Movement Malawi progress report.
 - Malawi joined SUN in **March 2011** (early riser). Since joining: stunting reduced from **47% to 33%**; iron deficiency from 51% to 22%; Vitamin A deficiency from 22% to 4%. [Scaling Up Nutrition](#)
- **Ruel-Bergeron, J.C. et al. (2021).** "Leveraging the SUN Movement to Operationalize Stunting Prevention Activities: Implementation Lessons From Rural Malawi." *Food & Nutrition Bulletin*. DOI: PubMed 34747237.
 - Ntchisi district stunting prevalence **40–50%** in children 6–23 months. "Right Foods at the Right Time" programme (2013–2018) included SQ-LNS and SBCC. [PubMed](#)

Methodological gap addressed: No prior study has used the 2024 MDHS for any ML analysis. Previous trend analyses rely on traditional regression methods unable to capture the complex non-linear interactions that explain 95% of the unexplained stunting decline (Kumchulesi, 2020).

2. Determinants of child stunting — empirical evidence base

2.1 Maternal anthropometry (height, weight, BMI)

- **Benjamin-Chung, J. et al. (2023).** "Causes and consequences of child growth faltering in low-resource settings." *Nature*, 621:568–576. DOI: 10.1038/s41586-023-06501-x.
 - **Landmark study.** 33 longitudinal cohorts (83,671 children, 662,763 measurements). Improving maternal anthropometry and child condition at birth accounted for LAZ increases of up to **0.40** by 24 months. **Pre-conception and pregnancy conditions are the single most impactful modifiable factor.**
- **Bawuah (2026).** "Malnourished Child, Overweight Mother?" *Maternal & Child Nutrition*. DOI: 10.1111/mcn.70175.
 - 103,497 mother–child pairs from 22 SSA countries. Maternal short stature: **AOR=3.00 (95% CI: 2.61–3.46)** — one of the strongest predictors identified across the literature. [Wiley Online Library](#)
- **Afolabi, S.A. & Palamuleni, M.E. (2021).** "Exploring determinants of under-5 stunting in Malawi using a GLMM." *South African Journal of Child Health*, 15(1). DOI: 10.7196/SAJCH.2021.v15i1.1756.
 - **Malawi-specific (MDHS 2015–16, n=5,686).** Overweight/obese mothers: **aOR=0.78 (95% CI: 0.65–0.94)**. Central region: 26% higher risk. Female sex: **aOR=0.83**. Secondary education: **aOR=0.73**. Wealth: **aOR=0.68**. [Journals](#) ~3% of variance at community level. [ResearchGate](#) [Journals](#)
 - **Relevance:** Directly applicable Malawi effect sizes. Confirms maternal BMI, education, wealth, sex, and region as essential ML features.

2.2 Child age and the first 1,000-day window

- **Victora, C.G. et al. (2010).** "Worldwide Timing of Growth Faltering: Revisiting Implications for Interventions." *Pediatrics*, 125(3):e473–e480.

- Seminal analysis. Growth faltering begins in utero and accelerates through first 24 months across all LMICs. The age-HAZ trajectory is universally consistent.
- **Ahsan, K.Z. et al. (2020).** "Determinants of stunting during the first 1000 days of life in Bangladesh." *Maternal & Child Nutrition*. DOI: PMC7500796.
 - Children 12–23 months: **2.65× more likely stunted** than 0–5 month-olds (aRR 2.65, 95% CI: 2.20–3.20). (PubMed Central) Mother's BMI <18.5, height <145 cm, age <18, and severe food insecurity all significant.
- **Martorell, R. (2017).** "Improved Nutrition in the First 1000 Days and Adult Human Capital and Health." *American Journal of Human Biology*. DOI: PMC5761352.
 - Stunting prevalence increases from ~30% at 0–5 months to ~55% by age 2. Largely irreversible after this period. (Wikipedia)
- **Systematic scoping review (2024).** "Prevalence and Determinants of Undernutrition Among Children 6 Months to 5 Years in SSA." *Nutrition Reviews*, Oxford Academic.
 - 59 publications. Most frequently reported determinants: child age >24 months (**n=23 studies**), male sex (**n=17**), maternal illiteracy (**n=11**), low household SES (**n=9**). (Oxford Academic)

2.3 Birth interval, birth order, and child sex

- **Zhang, Q. et al. (2022).** "Optimum birth interval (36–48 months) may reduce the risk of undernutrition in children: A meta-analysis." *Frontiers in Nutrition*.
 - 46 studies (898,860 children). Birth interval ≥24 months vs. <24 months: **OR=0.61 (95% CI: 0.55–0.67)** for stunting. Africa subgroup: **OR=0.53**. Optimal interval: 36–48 months. (Frontiers)
- **Wamani, H. et al. (2007).** "Boys are more stunted than girls in Sub-Saharan Africa: a meta-analysis of 16 DHS." *BMC Pediatrics*, 7:17. DOI: 10.1186/1471-2431-7-17.
 - Boys 40% stunted vs. girls 36%. **Adjusted OR=1.18 (95% CI: 1.14–1.22)**. Male disadvantage more pronounced in poorest quintiles. (Springer)
- **Garcia, S. et al. (2022).** "Greater male vulnerability to stunting?" *Annals of the New York Academy of Sciences*. DOI: PMC9205267.
 - Boys more likely born preterm, low birth weight, and at greater risk of respiratory disease, diarrhea, and malaria neonatally. Biological and immunological mechanisms proposed. (NCBI)

2.4 Maternal education and household wealth

- **Makoka, D. (2013).** "The Impact of Maternal Education on Child Nutrition: Evidence from Malawi, Tanzania and Zimbabwe." *DHS Working Papers No. 84*, ICF International.
 - **Malawi: 45% stunted.** Stunting significantly associated with maternal education ($p<0.0001$). Threshold: **≥10 years of schooling** for significant reduction. (Springer)
- **Akombi, B.J. et al. (2017).** "Stunting, Wasting and Underweight in Sub-Saharan Africa: A Systematic Review." *Int J Environ Res Public Health*, 14(8):863.
 - 49 SSA studies (1992–2017). Household SES, mother's education, child age, and sex were the **most consistently significant factors**.

2.5 Geographic and district-level determinants (Malawi)

- **Manda, S.O.M. et al. (2022).** "Shared geographic spatial risk of childhood undernutrition in Malawi." *Spatial and Spatio-temporal Epidemiology*. DOI: PMC9461579.
 - **Malawi-specific.** 5,066 children from MDHS 2015–16. Significant spatial clustering (Moran I=0.464, $p=0.009$). Southern districts followed by central at greatest risk.

- **Mtongwa, R.H. et al. (2025).** "Socio-economic and demographic determinants of undernutrition among 6–59 month old children living in Malawian stunting hotspots." *BMC Nutrition*. DOI: PMC11745028.
 - Three hotspot districts: Mzimba (38.9%), Mchinji (44%), Mangochi (45.4%).
 - [PubMed Central](#) Caregiver unemployment: **AOR=2.339 (95% CI: 1.340–4.081)**.
 - [PubMed Central](#)

2.6 UNICEF Conceptual Framework

- **UNICEF (2020).** "Conceptual Framework on the Determinants of Maternal and Child Nutrition." *UNICEF Nutrition Strategy 2020–2030*.
 - Updated framework: immediate determinants (diets, care); underlying determinants (food environments, feeding practices, health environments); enabling determinants (governance, resources, norms). [ResearchGate](#)
 - *Relevance:* Provides theoretical justification for hierarchical feature selection in ML models.
- **Black, R.E., Lutter, C.K. & Trude, A. (2020).** "All children surviving and thriving: re-envisioning UNICEF's conceptual framework." *Lancet Global Health*, 8(6):e766–e767.

Methodological gap addressed: Traditional regression studies identify linear, additive effects. ML methods capture the non-linear interactions and threshold effects (e.g., maternal height × wealth × child age) that the UNICEF framework implies but linear models cannot test.

3. Machine learning for child nutrition prediction — global literature

3.1 Meta-analysis establishing the evidence base

- **Rao, B., Rashid, M., Hasan, M.G. & Thunga, G. (2025).** "Machine Learning in Predicting Child Malnutrition: A Meta-Analysis of DHS Data." *Int J Environ Res Public Health*, 22(3):449. DOI: 10.3390/ijerph22030449.
 - **Systematic review + meta-analysis of 11 studies (2019–2023).** [PubMed Central](#) Pooled stunting accuracy: **68.92% (95% CI: 66.04–71.80)**. [PubMed Central](#) Pooled wasting accuracy: 84.39%. [MDPI +3](#) Gradient boosting: highest pooled sensitivity for stunting (**94.49%**). [ResearchGate](#) [PubMed Central](#) XGBoost: highest for wasting (88%). Bangladesh dominated (55% of studies); Africa 36%. [PubMed Central](#) [MDPI](#)
 - **Survey weights: NOT systematically addressed.** High heterogeneity ($I^2=100\%$). [MDPI](#)
 - *Relevance:* Establishes benchmark performance expectations. The proposed Malawi study can compare against these pooled estimates.

3.2 Malawi-specific ML studies (critical gap)

- **Mkungudza, J., Twabi, H.S. & Manda, S.O.M. (2024).** "Development of a diagnostic predictive model for determining child stunting in Malawi." *BMC Medical Research Methodology*, 24:175. DOI: 10.1186/s12874-024-02283-6.
 - **Only published study using Malawi DHS data for predictive stunting modeling.** Used MDHS 2015–16. Compared 7 variable selection methods for logistic regression. [Springer](#) [BMC Medical Research Methodology](#) **Best AUROC = 64% (95% CI: 60–67%)**. [UPSpace Repository](#) Urban AUROC=67%, rural=63%. Key predictors: wealth, child age, household size, birth type, birth weight.

- **Survey weights: NOT used. Only logistic regression, no advanced ML algorithms. Data from 2015–16.**
- *Gap addressed:* The proposed study extends this by using (1) 2024 MDHS, (2) multiple ML algorithms (RF, XGBoost, LightGBM), (3) survey weights, achieving potentially higher discrimination.
- **Chimimba et al. (2025).** "Advancing predictive analytics in child malnutrition: ML, ensemble and deep learning models." *Intelligent Systems with Applications*. DOI: 10.1016/j.iswa.2025.
 - Used **World Bank LSMS data for Malawi (NOT DHS)**. Seven algorithms. (ScienceDirect) RF: 100% wasting accuracy after SMOTE (overfitting concern). XGBoost: 95.52% stunting. DNN: 76.64% stunting. (ScienceDirect) **Survey weights: NO.** (ScienceDirect)
 - *Gap addressed:* Uses LSMS not DHS; inflated SMOTE accuracy; no weights. The proposed study uses DHS data with proper survey design.

3.3 Country-specific ML studies from Africa

- **Ndagijimana, S. et al. (2023).** "Prediction of Stunting Among Under-5 Children in Rwanda Using ML Techniques." *J Prev Med Public Health*. DOI: 10.3961/jpmph.22.388.
 - Rwanda DHS 2019–20, (Academia.edu) n=3,814. Gradient Boosting: **accuracy 79.33%**, **AUC 0.89**. Top predictors: mother's height, television, child's age, province, education. **Survey weights: NO.** (Jpmph)
- **Bitew, F.H., Sparks, C.S. & Nyarko, S.H. (2022).** "ML algorithms for predicting undernutrition in Ethiopia." *Public Health Nutrition*, 25(2):269–280. DOI: 10.1017/S1368980021003421.
 - 2016 EDHS, n=9,471. (PubMed Central) XGBoost highest accuracy: (PubMed Central) **67.7%**. Top predictors: time to water, anemia, child age, birth size, maternal underweight. (PubMed Central) **Survey weights: NO.** (Cambridge Core)
- **Ayele, M.K. et al. (2025).** "Predicting stunting status in Ethiopia using ensemble ML." *Scientific Reports*, 15:27907. DOI: 10.1038/s41598-025-03206-1.
 - EDHS 2011 & 2016 combined. RF: **accuracy 97.985%**, **ROC-AUC 99.995%** (after SMOTE). Multi-class classification. (Nature) **Survey weights: NO.** (Nature) Near-perfect accuracy likely inflated by SMOTE.
- **Chilyabanyama, O.N. et al. (2022).** "Performance of ML Classifiers for Stunting in Zambia." *Children*, 9(7):1082. DOI: 10.3390/children9071082.
 - 2018 ZDHS, n=6,799. RF: **79% test accuracy**. 13 of 58 features selected. Used calibration curves. (PubMed) (MDPI) **Survey weights: NO.** (PubMed)
- **Feleke, H.G. et al. (2026).** "ML for predicting acute malnutrition in East Africa." *Scientific Reports*. DOI: 10.1038/s41598-026-46944-6.
 - 12 East African countries, n=76,224. (Nature) RF best: **AUC 74.6%**. SHAP analysis identified rural residence, family size, birth size as top predictors. (Nature) **Survey weights: NO.**

3.4 ML studies from South Asia

- **Hasan, M.M. et al. (2024).** "Prediction of undernutrition using explainable ML in Bangladesh." *PLOS ONE*. DOI: 10.1371/journal.pone.0315393.
 - BDHS 2017–18. XGBoost best: **accuracy 81.73%**, **AUC 0.802** for stunting. (PLOS) Boruta feature selection + SHAP. **Survey weights: NO.**
- **Rahman, S.J. et al. (2021).** "Risk factors of stunting, wasting, and underweight in Bangladesh using ML." *PLOS ONE*, 16(6):e0253172. DOI: 10.1371/journal.pone.0253172.

- BDHS 2014, n=7,079. RF: **accuracy 88.3%** for stunting. (PubMed Central) **Survey weights: NO.**
- **Shen, H., Zhao, H. & Jiang, Y. (2023).** "ML for Predicting Stunting in Papua New Guinea." *Children*, 10(10):1638. DOI: 10.3390/children10101638.
 - PNG DHS 2016-18, n=3,380. LASSO-XGBoost: **AUC 0.767 (95% CI: 0.714-0.819).** (MDPI) (PubMed Central) SHAP dependence plots. Bayesian hyperparameter optimization. (PubMed Central) (MDPI) **Survey weights: NO.** (MDPI) (PubMed Central)

3.5 Feature importance methods (SHAP)

SHAP values are increasingly adopted for ML interpretability in nutrition studies. (DataCamp) Studies using SHAP include Shen et al. (2023), Hasan et al. (2024), Mussie et al. (2025 — Ethiopia adolescent girls), Feleke et al. (2026), and Islam et al. (2025 — Bangladesh). **Consistent top SHAP features across studies: child age, maternal height/BMI, household wealth, maternal education, birth size, and geographic region.** (PubMed Central +3)

3.6 Algorithm performance consensus

Across 22+ reviewed papers:

1. **Random Forest** — most frequently best performer (7+ studies)
2. **XGBoost** — best or competitive in 6+ studies; highest meta-analytic accuracy for wasting (PubMed Central)
3. **Gradient Boosting** — highest pooled sensitivity for stunting (94.49%) (PubMed Central)
4. **Logistic Regression** — competitive baseline; surprisingly strong in some studies
5. **LightGBM/CatBoost** — emerging alternatives with competitive performance

Critical finding: Of 22 papers reviewed, only 1 (Alam et al., 2021, Bangladesh) explicitly incorporated survey weights into ML models. This represents a ~95% gap rate — the central methodological innovation the proposed study addresses.

4. DHS data methodology and survey weights in ML — the critical gap

4.1 Evidence that ignoring weights biases ML results

- **MacNeill, N., Foynes, M.M. & Fisher, L. (2023).** "Implementing ML methods with complex survey data: Lessons learned on the impacts of accounting sampling weights in gradient boosting." *PLOS ONE*, 18(1):e0280387.
 - **THE KEY METHODOLOGICAL PAPER.** Using NHANES III (N=15,820), unweighted model F1: **81.9%** vs. weighted F1: **77.4%** — a **4.5 percentage point inflation.** (PubMed Central) Post-hoc recalculation partially mitigated error (F1: 77.0%). For N<5,000 with high weight variability, ignoring weights produced inconsistent results. (PubMed Central) Quote: "Few ML software implementations offer options for handling complex samples... a major challenge impeding broader incorporation of ML into epidemiologic research." (ResearchGate) (PubMed Central)
 - **Relevance: Directly demonstrates** that ignoring DHS survey weights inflates performance estimates. Primary citation for justifying the study's survey-weighted approach.
- **Toth, D. & Eltinge, J.L. (2011).** "Building Consistent Regression Trees from Complex Sample Data." *Journal of the American Statistical Association*, 106(496):1626-1636.
 - **Foundational paper.** Proposed incorporating inverse selection probabilities into tree splitting criteria. Weighted estimator was **substantially less biased** than unweighted. (PubMed Central) Quote: "The use of [recursive partitioning] in survey analysis has

largely ignored design information such as the selection probabilities."

U.S. Bureau of Labor Statistics

- **Danieli, C. & Brunner, M. (2024).** "Learning de-biased regression trees and forests from complex samples." *Machine Learning*, Springer.
 - Most recent advance. Proposes de-biasing methods bridging tree learning to survey variance estimation. Quote: "'Naive estimation' that simply ignores the sampling weights may be substantially biased." [Springer](#)

4.2 Cross-validation for complex survey data

- **Wieczorek, J., Guerin, C. & McMahon, T. (2022).** "K-Fold Cross-Validation for Complex Sample Surveys." *Stat*, 11(1):e454.
 - **Critical paper.** Standard k-fold CV assumes exchangeability, which **does not hold** for DHS-type designs. Proposes "Survey CV": all observations from a cluster must be in the same fold. [Wiley Online Library](#) Ignoring clustering leads to **underestimated prediction error** and overfitting. [Rbind](#)
 - **Relevance:** Directly justifies cluster-aware cross-validation in the Malawi study.
- **Kern, C., Klausch, T. & Kreuter, F. (2020).** "Tree-based ML Methods for Survey Research." *Survey Research Methods*, 13(1):73–93. DOI: PMC7425836.
 - Comprehensive review of CART, random forests, boosting, and BART in survey contexts. Discusses weight incorporation for nonresponse adjustment and prediction. [PubMed Central](#)
- **Lumley, T. (2024).** "Crossvalidation in complex survey data." Blog post.
 - Describes replicate-weight-based approach. Quote: "Using a cross-validation split that ignores clustering tended to choose a LASSO regularisation parameter that was smaller than ideal, resulting in overfitting." [Rbind](#)

4.3 DHS sampling methodology references

- **Croft, T.N., Marshall, A.M.J., Allen, C.K. et al. (2018).** *Guide to DHS Statistics (DHS-7)*. Rockville, MD: ICF. [Updated 2023 as DHS-8]
 - DHS uses **two-stage stratified probability sampling**: Stage 1 — census EAs selected with PPS within strata; Stage 2 — fixed household count by equal probability systematic sampling. [GovInfo](#) [PLOS](#) Weight variable: **v005** (divided by 1,000,000). Complex sample parameters: PSU (v021), stratification (v023), weight (v005). [The DHS Program](#)
- **Elkasabi, M. (2020).** "Sampling Design in DHS Surveys." *DHS Working Papers No. 176*.
- **Aliaga, A. & Ren, R. (2006).** "Optimal Sample Sizes for Two-stage Cluster Sampling in DHS." *DHS Working Papers No. 30*. [The DHS Program](#)

4.4 Technical implementation

- **XGBoost:** Natively supports instance weights via `sample_weight` parameter. Weights modify gradient and hessian calculations during training.
- **LightGBM:** Supports `weight` column in Dataset constructor.
- **Random Forest (scikit-learn):** Supports `sample_weight` in `fit()`.
- **rpms R package** (Toth, D.): Purpose-built for design-consistent trees and random forests with complex survey data. Functions: `rpms()`, `rpms_forest()`, `rpms_boost()`.
- **surveyCV R package** (Wieczorek et al.): Implements design-based CV fold creation compatible with `survey` package.
- **Lumley, T. (2004).** "Analysis of Complex Survey Samples." *Journal of Statistical Software*, 9(8):1–19. And: **Lumley, T. (2010).** *Complex Surveys: A Guide to Analysis Using R*. Wiley.

4.5 The gap documented

Study	Country	DHS data	Survey weights in ML?
Mkungudza et al. (2024)	Malawi	MDHS 2015-16	NO
Chimimba et al. (2025)	Malawi	LSMS	NO
Bitew et al. (2022)	Ethiopia	EDHS 2016	NO
Ayele et al. (2025)	Ethiopia	EDHS 2011/2016	NO
Ndagijimana et al. (2023)	Rwanda	RDHS 2019-20	NO
Chilyabanyama et al. (2022)	Zambia	ZDHS 2018	NO
Rahman et al. (2021)	Bangladesh	BDHS 2014	NO
Hasan et al. (2024)	Bangladesh	BDHS 2017-18	NO
Shen et al. (2023)	PNG	PNG DHS 2016-18	NO
Feleke et al. (2026)	East Africa (12 countries)	Multiple DHS	NO
Alam et al. (2021)	Bangladesh	BDHS 2014	YES

Alam, M.J. et al. (2021). "Sociodemographic risk factors of under-five stunting in Bangladesh: Assessing the role of interactions using a ML method." *PLOS ONE*. DOI: PMC8407547.

- Quote: "Survey weights are duly applied in both the classification tree and logistic regression models." — **the sole exception** among 20+ reviewed studies.

Methodological gap addressed: The proposed study is among the first globally to systematically integrate DHS survey weights into multi-algorithm ML prediction, using cluster-aware cross-validation.

5. Spatial analysis of stunting

5.1 Large-scale geospatial mapping

- **Local Burden of Disease Child Growth Failure Collaborators (2020).** "Mapping child growth failure across low- and middle-income countries." *Nature*, 577(7789):231–234. DOI: 10.1038/s41586-019-1878-8.
 - Bayesian model-based geostatistics with georeferenced data, producing **5×5 km resolution** estimates across 105 LMICs (2000–2017). Revealed striking subnational heterogeneity masked by national statistics.
- **Local Burden of Disease Collaborators (2018).** "Mapping child growth failure in Africa between 2000 and 2015." *Nature*, 555(7694):41–47. DOI: 10.1038/nature25760.
 - First high-resolution (5×5 km) geospatial CGF estimates for Africa. Demonstrates that geospatial estimates enable precision public health targeting.

5.2 Bayesian geostatistical models

- **Uwiringiyimana, V. et al. (2022).** "Bayesian geostatistical modelling of stunting in Rwanda." *BMC Public Health*, 22:159. DOI: 10.1186/s12889-022-12552-y.
 - Rwanda DHS 2015; 5×5 km predictions. Spatial residual mapping revealed elevated stunting even after covariate adjustment.

- **Ahmed, K.Y. et al. (2021).** "Mapping Geographical Differences and Determinants of Childhood Stunting in Ethiopia." *Nutrients*, 13(6):2104. DOI: 10.3390/nu13062104.
 - EDHS 2016; Bayesian INLA model. Prevalence ranged **25.5% to 56.6%** across zones. Incorporated climatic covariates (aridity, temperature, rainfall).
- **Aheto, J.M.K. & Dagne, G.A. (2021).** "Geostatistical analysis and environmental determinants of under-5 stunting." *The Lancet Planetary Health*, 5(6):e347–e355. DOI: 10.1016/S2542-5196(21)00080-2.
 - Ghana DHS; Bayesian INLA model. **Elevation, precipitation, and aridity** were significant environmental predictors. Web-based interactive maps with exceedance probabilities.

5.3 Malawi-specific spatial studies

- **Ngwira, A. (2022).** "Shared geographic spatial risk of childhood undernutrition in Malawi." *Public Health in Practice*, 3:100224. DOI: 10.1016/j.puhip.2021.100224.
 - MDHS 2015–16; 5,066 children. **Moran's I=0.464 (p=0.009)** for stunting-wasting shared risk. Southern districts greatest risk. Climate (high rainfall SE, high temperature SW) identified as spatial driver.

5.4 Environmental covariates and child health

- **Tusting, L.S. et al. (2020).** "Environmental temperature and growth faltering in African children." *The Lancet Planetary Health*, 4(12):e540–e550. DOI: 10.1016/S2542-5196(20)30037-1.
 - **656,107 children** from 52 DHS surveys linked with remotely sensed LST, EVI, precipitation, nighttime lights. Monthly mean daytime LST >35°C: wasting **OR=1.27**, concurrent stunting-wasting **OR=1.23**.
- **Shively, G. (2022).** "Altitude and early child growth in 47 countries." *Population and Environment*, 43:390–409. DOI: 10.1007/s11111-021-00390-w.
 - 630,499 children across 47 countries. Per 1,000m elevation gain, HAZ declined by **0.195 points**, after controlling for household, community, and environmental factors.
- **Baffour, B. et al. (2023).** "Geostatistical modelling of child undernutrition using remote-sensed data." *Scientific Reports*, 13:21573. DOI: 10.1038/s41598-023-48980-y.
 - Bangladesh and Ghana DHS supplemented by remote-sensing covariates (aridity, vegetation, climate). Demonstrated that harnessing remotely sensed data improves geographical stunting estimates.

5.5 Spatial ML and explainable AI

- **Singh, M. et al. (2025).** "Multidimensional Correlates of Childhood Stunting in India: Spatial ML and Explainable AI." *Challenges*, 9(2):34. DOI: 10.3390/challe9020034.
 - Integrated spatial econometric models (spatial lag, spatial error) with ML (RF, XGBoost) and SHAP. **Moran's I=0.520** — pronounced spatial spillover. ML captured nonlinear relationships beyond conventional spatial regression.
 - **Relevance: Directly relevant methodological template** for combining spatial analysis with ML + SHAP for stunting prediction.

5.6 DHS GPS displacement methodology

- **Burgert, C.R. et al. (2013).** "Geographic Displacement Procedure and Georeferenced Data Release Policy for DHS." *DHS Spatial Analysis Reports No. 7*.
 - Urban clusters displaced **0–2 km**; rural clusters **0–5 km** (1% up to 10 km). Restricted to within country boundaries and DHS survey region.

- **Perez-Heydrich, C. et al. (2016).** "Influence of DHS Point Displacements on Distance-Based Analyses." *Spatial Demography*. DOI: PMC4946438.
 - Displacement introduces measurement error; proposes regression calibration. Contextual variables should be calculated within **5–10 km buffers**.

Key environmental covariates documented across studies: NDVI/EVI (vegetation), precipitation and temperature (means and anomalies), elevation/altitude, aridity index, land surface temperature, nighttime lights (urbanization proxy), distance to roads/health facilities.

6. Calibration and model evaluation in health ML

6.1 Why calibration matters for stunting prediction

- **Van Calster, B. et al. (2019).** "Calibration: the Achilles heel of predictive analytics." *BMC Medicine*, 17:230.
 - **Landmark paper.** Calibration is systematically neglected. Quote: "*Poorly calibrated algorithms can be misleading and potentially harmful for clinical decision-making.*" Defines calibration-in-the-large, calibration slope, and calibration curves.
 - **Relevance:** Primary justification for including calibration assessment. If stunting models inform resource allocation, miscalibrated probabilities misallocate interventions.
- **Van Calster, B. & Vickers, A.J. (2015).** "Calibration of Risk Prediction Models: Impact on Decision-Analytic Performance." *Med Decis Making*, 35(2):162–169.
 - Even well-discriminating models can cause harm if miscalibrated.
- **Christodoulou, E. et al. (2019).** "A systematic review shows no performance benefit of ML over logistic regression for clinical prediction models." *J Clin Epidemiol*, 110:12–22.
 - 71 studies reviewed. Median AUC difference between ML and LR: minimal. **79% did not assess calibration.** 68% had potential bias in validation.
 - **Relevance:** Justifies including LR as benchmark and reporting calibration (which most ML studies omit).

6.2 Brier score and overall accuracy

- **Steyerberg, E.W. et al. (2010).** "Assessing the performance of prediction models." *Epidemiology*, 21(1):128–138.
 - Brier score is the **mean squared error of predicted risks** — an overall measure comprising both discrimination and calibration. Recommends reporting separately alongside Brier.
- **Huang, Y. et al. (2020).** "A tutorial on calibration measurements and calibration models for clinical prediction models." *JAMIA*, 27(4):621–633.
 - Low Brier score does not guarantee good calibration — must complement with calibration-specific metrics. Covers Platt scaling, isotonic regression.

6.3 AUC comparison methods

- **DeLong, E.R., DeLong, D.M. & Clarke-Pearson, D.L. (1988).** "Comparing the areas under two or more correlated ROC curves: a nonparametric approach." *Biometrics*, 44(3):837–845.
 - Original nonparametric test for comparing correlated AUCs using generalized U-statistics.
 - **Relevance:** Standard method for comparing RF vs. XGBoost vs. LightGBM vs. LR AUCs.

- Demler, O.V. et al. (2012). "Misuse of DeLong test to compare AUCs for nested models." *Stat Med*, 31(23):2572–2587.
 - DeLong should NOT be used for nested models. Appropriate for comparing different algorithms on same features.

6.4 Post-hoc calibration methods

- Niculescu-Mizil, A. & Caruana, R. (2005). "Predicting good probabilities with supervised learning." *ICML 2005*.
 - **Boosted trees and SVMs benefit most from calibration.** Isotonic regression outperforms Platt scaling when $\geq 1,000$ calibration samples. LR and neural nets already well-calibrated.
 - *Relevance:* With DHS sample sizes ($>5,000$), isotonic regression is appropriate for calibrating XGBoost/gradient boosting stunting models.
- Platt, J.C. (2000). "Probabilistic outputs for SVMs." *Advances in Large Margin Classifiers*, MIT Press, pp. 61–74.

6.5 Subgroup analysis and fairness

- Chen, R.J. et al. (2023). "Algorithm fairness in AI for medicine and healthcare." *Nature Machine Intelligence*, 5:175–187.
 - Quote: "If left unchecked, algorithms may amplify existing healthcare inequities." Justifies evaluating model performance across age, region, sex, and wealth subgroups.

6.6 Performance ceiling of DHS-based models

The meta-analysis by Rao et al. (2025) establishes realistic performance expectations: **pooled AUC ~0.69 for stunting** using DHS data. Best individual studies achieve AUCs of 0.77–0.89. This ceiling reflects inherent DHS limitations: no dietary intake data, no biomarkers in most modules, cross-sectional design, and anthropometric measurement error. **Studies reporting AUC >0.95 with SMOTE should be viewed with caution** (possible data leakage/overfitting).

6.7 Reporting guidelines

- Collins, G.S. et al. (2024). "TRIPOD+ AI statement: updated guidance for reporting clinical prediction models that use regression or ML." *BMJ*, 385:e078378.
 - 27-item checklist superseding TRIPOD 2015. Specifically designed for ML models.
- Steyerberg, E.W. (2019). *Clinical Prediction Models: A Practical Approach*. 2nd ed. Springer.
 - Definitive textbook. ABCD validation framework.
- Alba, A.C. et al. (2017). "Discrimination and Calibration of Clinical Prediction Models." *JAMA*, 318(14):1377–1384.
 - AUC >0.75 indicates good discrimination. A useful model needs both satisfactory discrimination and calibration.

7. Malawi-specific policy and programme context

7.1 Health system

Malawi's health system operates on three tiers with a **severe workforce crisis: 1 doctor per 88,300 people** and a 45% health worker vacancy rate. Health Surveillance Assistants (HSAs) are the backbone of community nutrition services, conducting growth monitoring, CMAM, and iCCM. Malawi was a **global CMAM pioneer**, scaling nationally to all 29 districts by 2006. The

HSSP III (2023–2030) emphasizes digital health and decentralized information systems, creating policy demand for ML-based surveillance tools.

7.2 Agricultural subsidies and food security

- **Walls, H. et al. (2023).** "Exploring the agriculture-nutrition policy nexus in Malawi." *PLOS Global Public Health*. DOI: PMC10566744.
 - FISP (2005/06–2020) provided subsidized fertilizer and seeds to ~1.5 million families, consuming **40–70% of agriculture ministry budget**. Replaced by AIP in 2020, which removed legume subsidies, potentially reinforcing maize monoculture at the expense of dietary diversity.
- **2024 El Niño crisis context:** President Chakwera declared State of Disaster in **23 of 28 districts** (March 2024). **4.2 million people** in IPC Phase 3+ food insecurity (May–September 2024), rising to **5.7 million (28% of population)** during October 2024–March 2025. SAM admissions increased **18%**. Inflation ~38.8%.
 - *Relevance:* The 2024 MDHS data were collected during this compounding crisis, creating unique conditions for models that identify the most vulnerable children during emergencies.

7.3 Climate-health projects

- **GCF CHWBRC project (FP244).** Approved October 2024 by Green Climate Fund. **\$37 million** (\$33M GCF grant + co-financing from GSK, Sanofi, FCDO). Implemented by Save the Children across **6 southern Malawi districts**. Targets 1.7 million people over 5 years (2025–2029). Focus: climate-resilient health infrastructure, Health Early Warning and Response System (HEWARS), ICCM, and maternal/child nutrition.
 - *Relevance:* Climate-informed ML prediction models directly support CHWBRC's surveillance objectives. Environmental covariates (temperature, precipitation, NDVI) can be integrated as spatial features.

7.4 Malawi Vision 2063 and strategic alignment

MW2063 (launched 2020) aspires to upper-middle-income status by 2063. **Enabler 5** pledges access to nutritious food during the first 1,000 days. The **National Multi-Sector Nutrition Strategic Plan 2025–2030**, aligned with SUN 3.0, explicitly calls for improved nutrition information systems and evidence-based targeting — creating direct operational demand for ML prediction models.

7.5 Precision public health targeting

- **Kenya AI malnutrition forecasting (2025).** USC/Microsoft AI for Good/Amref/Kenya MOH. ML model integrating DHIS2 clinical data + satellite crop data predicts acute malnutrition **up to 6 months ahead with 89% accuracy**. Dashboard for regional risk visualization.
 - *Relevance:* Demonstrates the operational template for translating ML stunting prediction into real-time surveillance in African health systems. The proposed Malawi study could inform a similar system.

Summary of research gaps the proposed study addresses

The literature reveals five interlocking gaps that the proposed study fills:

1. **No survey-weighted ML for stunting prediction.** Of 22+ ML studies using DHS data reviewed, only one (Alam et al., 2021) incorporated survey weights. MacNeill et al. (2023)

demonstrated this inflates performance by ~4.5 percentage points. The proposed study systematically integrates v005 weights into model training.

2. **No advanced ML applied to Malawi DHS data.** Mkungudza et al. (2024) used only logistic regression on MDHS 2015–16 (AUC=0.64). Chimimba et al. (2025) used LSMS, not DHS. The proposed study is the first to apply RF, XGBoost, and LightGBM to any Malawi DHS round.
3. **No study uses 2024 MDHS data.** As the most recent nationally representative dataset (collected during the 2024 El Niño crisis), it captures current risk factor distributions unavailable in any prior analysis.
4. **Calibration systematically neglected.** Christodoulou et al. (2019) found 79% of ML prediction studies omit calibration. The proposed study reports Brier scores, calibration curves, and tests calibration across subgroups — essential for public health resource allocation.
5. **Cluster-aware cross-validation absent.** Wiecek et al. (2022) showed standard k-fold CV produces overly optimistic estimates with clustered survey data. The proposed study uses cluster-level fold assignment to produce honest performance estimates.

These gaps position the study as a methodological contribution to the broader field of ML-based health prediction with complex survey data, while producing the first nationally representative, calibrated stunting risk scores for Malawi's 28 districts.